Homework 12

Zhiqi Gan

Problem 12.1.
Solution 12.1.
1. eigenvalues $\lambda_1 = \lambda_2 = \lambda_3 = 1$, eigenvectors $\text{span}(\boldsymbol{e}_1)$.

2. eigenvalues and eigenvectors: $\lambda_1 = 1, \text{span}(\begin{bmatrix}1\\0\\0\end{bmatrix})$. $\lambda_2 = 2, \text{span}(\begin{bmatrix}1\\1\\0\end{bmatrix})$. $\lambda_3 = 3, \text{span}(\begin{bmatrix}1\\2\\2\end{bmatrix})$.

3. eigenvalues and eigenvectors: $\lambda_1 = 7, \text{span}(\begin{bmatrix}-11\\3\end{bmatrix})$. $\lambda_2 = 21, \text{span}(\begin{bmatrix}1\\1\end{bmatrix})$.

4. eigenvalues and eigenvectors: $\lambda_1 = \lambda_2 = 1, \text{span}(\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix})$. $\lambda_3 = -1, \text{span}(\begin{bmatrix}-1\\0\\1\end{bmatrix})$.

5. Note that $\text{trace}(A) = 2b, \det(A) = a^2 + b^2$. So $\lambda_1 = b + ia, \lambda_2 = b - ia$, the corresponding eigenvectors are $\text{span}(\begin{bmatrix}i\\1\end{bmatrix})$ and $\text{span}(\begin{bmatrix}1\\i\end{bmatrix})$.

6. eigenvalues and eigenvectors: $\lambda_1 = 0, \text{span}(\begin{bmatrix}2\\-2\\1\end{bmatrix})$. $\lambda_2 = 3i, \text{span}(\begin{bmatrix}-1-3i\\1-3i\\4\end{bmatrix})$. $\lambda_2 = -3i, \text{span}(\begin{bmatrix}-1+3i\\1+3i\\4\end{bmatrix})$.

Problem 12.2.
Solution 12.2.
1.
$$\begin{bmatrix}a_{n+2}\\a_{n+1}\end{bmatrix} = \begin{bmatrix}3 & -2\\1 & 0\end{bmatrix}\begin{bmatrix}a_{n+1}\\a_n\end{bmatrix} \Longrightarrow \begin{bmatrix}a_{n+1}\\a_n\end{bmatrix} = \begin{bmatrix}3 & -2\\1 & 0\end{bmatrix}^n \begin{bmatrix}a_1\\a_0\end{bmatrix}$$

To calculate $\begin{bmatrix}3 & -2\\1 & 0\end{bmatrix}^n$, we need to diagonalize it. It's obvious to see that $\lambda_1 = 1$, corresponding eigenspace is $\text{span}(\begin{bmatrix}1\\1\end{bmatrix})$, $\lambda_2 = 2$, corresponding eigenspace is $\text{span}(\begin{bmatrix}2\\1\end{bmatrix})$, then,

$$A = \begin{bmatrix}3 & -2\\1 & 0\end{bmatrix} = X\Lambda X^{-1} = \begin{bmatrix}1 & 2\\1 & 1\end{bmatrix}\begin{bmatrix}1 & \\ & 2\end{bmatrix}\begin{bmatrix}-1 & 2\\1 & -1\end{bmatrix}$$

Hence,
$$a_n = \boldsymbol{e}_2^{\mathrm{T}} A^n \begin{bmatrix}a_1\\a_0\end{bmatrix} = \boldsymbol{e}_2^{\mathrm{T}} \begin{bmatrix}1 & 2\\1 & 1\end{bmatrix}\begin{bmatrix}1 & \\ & 2^n\end{bmatrix}\begin{bmatrix}-1 & 2\\1 & -1\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix} = 2^n - 1.$$

2.
$$\begin{bmatrix}a_{n+1}\\1\end{bmatrix} = \begin{bmatrix}2 & 1\\0 & 1\end{bmatrix}\begin{bmatrix}a_n\\1\end{bmatrix} \Longrightarrow \begin{bmatrix}a_n\\1\end{bmatrix} = \begin{bmatrix}2 & 1\\0 & 1\end{bmatrix}^n \begin{bmatrix}a_0\\1\end{bmatrix}$$

Similar to (1), after diagonalization, we have $a_n = 2^n - 1$.

3.
$$\begin{bmatrix}a_{n+1}\\2^{n+1}\end{bmatrix} = \begin{bmatrix}2 & 1\\0 & 2\end{bmatrix}\begin{bmatrix}a_n\\2^n\end{bmatrix} \Longrightarrow \begin{bmatrix}a_n\\2^n\end{bmatrix} = \begin{bmatrix}2 & 1\\0 & 2\end{bmatrix}^n \begin{bmatrix}a_0\\2^0\end{bmatrix}$$

$\begin{bmatrix}2 & 1\\0 & 2\end{bmatrix}$ is not diagonalizable, but

$$\begin{bmatrix}2 & 1\\0 & 2\end{bmatrix}^n = 2^n \begin{bmatrix}1 & \frac{1}{2}\\0 & 1\end{bmatrix}^n = 2^n \begin{bmatrix}1 & \frac{n}{2}\\0 & 1\end{bmatrix} = \begin{bmatrix}2^n & \frac{n}{2}2^n\\0 & 2^n\end{bmatrix}$$

Hence
$$a_n = \boldsymbol{e}_1^{\mathrm{T}} \begin{bmatrix}a_n\\2^n\end{bmatrix} = \boldsymbol{e}_1^{\mathrm{T}} \begin{bmatrix}2 & 1\\0 & 2\end{bmatrix}^n \begin{bmatrix}a_0\\2^0\end{bmatrix} = \boldsymbol{e}_1^{\mathrm{T}} \begin{bmatrix}2^n & \frac{n}{2}2^n\\0 & 2^n\end{bmatrix}\begin{bmatrix}a_0\\2^0\end{bmatrix} = \frac{n}{2}2^n = n2^{n-1}.$$

4.
$$\begin{bmatrix}a_{n+1}\\b_{n+1}\end{bmatrix} = \begin{bmatrix}1 & 1\\-2 & 4\end{bmatrix}\begin{bmatrix}a_n\\b_n\end{bmatrix} \Longrightarrow \begin{bmatrix}a_n\\b_n\end{bmatrix} = \begin{bmatrix}1 & 1\\-2 & 4\end{bmatrix}^n \begin{bmatrix}a_0\\b_0\end{bmatrix}$$

Similar to (1), after diagonalization, we have $a_n = 3^n - 2^n$.

5.
$$\begin{bmatrix}a_{n+2}\\a_{n+1}\end{bmatrix} = \begin{bmatrix}2 & -1\\1 & 0\end{bmatrix}\begin{bmatrix}a_{n+1}\\a_n\end{bmatrix} \Longrightarrow \begin{bmatrix}a_{n+1}\\a_n\end{bmatrix} = \begin{bmatrix}2 & -1\\1 & 0\end{bmatrix}^n \begin{bmatrix}a_1\\a_0\end{bmatrix}$$

$\begin{bmatrix}2 & -1\\ 1 & 0\end{bmatrix}$ is not diagonalizable, but we can do Jordan decomposition,

$$\begin{bmatrix}2 & -1\\ 1 & 0\end{bmatrix} = \begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\begin{bmatrix}0 & 1\\ 1 & -1\end{bmatrix}$$

So

$$\begin{bmatrix}2 & -1\\ 1 & 0\end{bmatrix}^n = \begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}^n\begin{bmatrix}0 & 1\\ 1 & -1\end{bmatrix} = \begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}1 & n\\ 0 & 1\end{bmatrix}\begin{bmatrix}0 & 1\\ 1 & -1\end{bmatrix}$$

Then

$$a_n = \boldsymbol{e}_2^{\mathrm{T}}\begin{bmatrix}2 & -1\\ 1 & 0\end{bmatrix}^n\begin{bmatrix}a_1\\ a_0\end{bmatrix} = \boldsymbol{e}_2^{\mathrm{T}}\begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}1 & n\\ 0 & 1\end{bmatrix}\begin{bmatrix}0 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}a_1\\ a_0\end{bmatrix} = n.$$

6.

$$\begin{bmatrix}a_{n+1}\\ (n+1)^2\\ n+1\\ 1\end{bmatrix} = \begin{bmatrix}1 & 1 & 0 & 0\\ 0 & 1 & 2 & 1\\ 0 & 0 & 1 & 1\\ 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}a_n\\ n^2\\ n\\ 1\end{bmatrix} \implies \begin{bmatrix}a_n\\ n^2\\ n\\ 1\end{bmatrix} = \begin{bmatrix}1 & 1 & 0 & 0\\ 0 & 1 & 2 & 1\\ 0 & 0 & 1 & 1\\ 0 & 0 & 0 & 1\end{bmatrix}^n\begin{bmatrix}a_0\\ 0\\ 0\\ 1\end{bmatrix}$$

Set $A = \begin{bmatrix}1 & 1 & 0 & 0\\ 0 & 1 & 2 & 1\\ 0 & 0 & 1 & 1\\ 0 & 0 & 0 & 1\end{bmatrix} = I + X$, then $X^2 = \begin{bmatrix}0 & 0 & 2 & 1\\ 0 & 0 & 0 & 2\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}$, $X^3 = \begin{bmatrix}0 & 0 & 0 & 2\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}$, $X^k = O, k > 3$. So

$$A^n = (I+X)^n = I + C_n^1 X + C_n^2 X^2 + C_n^3 X^3$$

Hence

$$a_n = \boldsymbol{e}_1^{\mathrm{T}} A^n \begin{bmatrix}a_0\\ 0\\ 0\\ 1\end{bmatrix} = \boldsymbol{e}_1^{\mathrm{T}}(I + C_n^1 X + C_n^2 X^2 + C_n^3 X^3)\boldsymbol{e}_4 = C_n^2 + 2C_n^3$$

Problem 12.3.
Solution 12.3.
1.

$$\det(\lambda I - A) = \begin{vmatrix}\lambda - A & -b\\ -c & \lambda - d\end{vmatrix} = \lambda^2 - (a+d)\lambda + ad - bc$$

So the two eigenvalues are $\lambda = \frac{a+d\pm\sqrt{(a+d)^2-4ad+4bc}}{2}$.
2.

$$A\begin{bmatrix}b\\ \lambda - a\end{bmatrix} = \lambda\begin{bmatrix}b\\ \lambda - a\end{bmatrix} \iff (A - \lambda I)\begin{bmatrix}b\\ \lambda - a\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$$

Thus,

$$\begin{bmatrix}a-\lambda & b\\ c & d-\lambda\end{bmatrix}\begin{bmatrix}b\\ \lambda - a\end{bmatrix} = \begin{bmatrix}0\\ -\lambda^2 + (a+d)\lambda - ad + bc\end{bmatrix}$$

λ is an eigenvalue of A, so $\lambda^2 - (a+d)\lambda + ad - bc = 0 \implies \begin{bmatrix}a-\lambda & b\\ c & d-\lambda\end{bmatrix}\begin{bmatrix}b\\ \lambda - a\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix} \implies A\begin{bmatrix}b\\ \lambda - a\end{bmatrix} = \lambda\begin{bmatrix}b\\ \lambda - a\end{bmatrix}$.
Similarly, $A\begin{bmatrix}\lambda - d\\ c\end{bmatrix} = \lambda\begin{bmatrix}\lambda - d\\ c\end{bmatrix}$.
3.

$$\begin{bmatrix}b\\ \lambda - a\end{bmatrix} = \begin{bmatrix}\lambda - d\\ c\end{bmatrix} = \mathbf{0} \implies b = c = 0, \lambda - d = \lambda - a = 0 \implies A = \begin{bmatrix}\lambda & \\ & \lambda\end{bmatrix}$$

Problem 12.4.
Solution 12.4.
1. Proof:
$A\boldsymbol{v} = \lambda\boldsymbol{v} \implies A^2\boldsymbol{v} = \lambda^2\boldsymbol{v} \implies \cdots \implies A^n\boldsymbol{v} = \lambda^n\boldsymbol{v}$. Set $p(x) = k_n x^n + k_{n-1}x^{n-1} + \cdots + k_0$, then

$$\begin{aligned}p(A)\boldsymbol{v} &= (k_n A^n + k_{n-1}A^{n-1} + \cdots + k_0 I)\boldsymbol{v}\\ &= k_n A^n\boldsymbol{v} + k_{n-1}A^{n-1}\boldsymbol{v} + \cdots + k_0\boldsymbol{v}\\ &= k_n\lambda^n\boldsymbol{v} + k_{n-1}\lambda^{n-1}\boldsymbol{v} + \cdots + k_0\boldsymbol{v}\\ &= (k_n\lambda^n + k_{n-1}\lambda^{n-1} + \cdots + k_0)\boldsymbol{v}\\ &= p(\lambda)\boldsymbol{v}\end{aligned}$$

2. $p(A) = (A-I)^2 + I = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 2 \end{bmatrix}$. The eigenvalues of $p(A)$ are $\lambda_1 = \lambda_2 = 1, \lambda_3 = 2$. Obviously, $\boldsymbol{e}_2$ is an eigenvector of $p(A)$ but not an eigenvector of A.

3. $p(A) = (A-I)(A-2I) + I = \begin{bmatrix} 1 & -1 & \\ & 1 & \\ & & 1 \end{bmatrix}$. The eigenvalues of $p(A)$ are $\lambda_1 = \lambda_2 = \lambda_3 = 1$, and the eigenvectors of $p(A)$ are $\operatorname{span}(\boldsymbol{e}_1, \boldsymbol{e}_3)$. But the eigenvectors of A are only $\operatorname{span}(\boldsymbol{e}_1)$ and $\operatorname{span}(\boldsymbol{e}_3)$. So $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ is an eigenvector of $p(A)$ but not an eigenvector of A.

Problem 12.5.

Solution 12.5.

1.

$$A = \begin{bmatrix} 4 & -1 & -1 & -1 \\ -1 & 4 & -1 & -1 \\ -1 & -1 & 4 & -1 \\ -1 & -1 & -1 & 4 \end{bmatrix}, H_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

Then,

$$AH_4 = \begin{bmatrix} 4 & -1 & -1 & -1 \\ -1 & 4 & -1 & -1 \\ -1 & -1 & 4 & -1 \\ -1 & -1 & -1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 5 & 5 & 5 \\ 1 & -5 & 5 & -5 \\ 1 & 5 & -5 & -5 \\ 1 & -5 & -5 & 5 \end{bmatrix}.$$

2. From 1, we have

$$AH_4 = \begin{bmatrix} 1 & 5 & 5 & 5 \\ 1 & -5 & 5 & -5 \\ 1 & 5 & -5 & -5 \\ 1 & -5 & -5 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 5 & & \\ & & 5 & \\ & & & 5 \end{bmatrix} = H_4 D$$

$$H_4^{-1} A H_4 = D = \begin{bmatrix} 1 & & & \\ & 5 & & \\ & & 5 & \\ & & & 5 \end{bmatrix}$$

3. $\det(A) = 5^3 = 125$. $\quad A^{-1} = H_4 \begin{bmatrix} 1 & & & \\ & \frac{1}{5} & & \\ & & \frac{1}{5} & \\ & & & \frac{1}{5} \end{bmatrix} H_4^{-1} = \frac{1}{5} \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{bmatrix}$.

Problem 12.6.

Solution 12.6.

1.

$$AF_4 = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix} = \begin{bmatrix} 10 & -2-2i & -2 & -2+2i \\ 10 & 2-2i & 2 & 2+2i \\ 10 & 2+2i & -2 & 2-2i \\ 10 & -2+2i & 2 & -2-2i \end{bmatrix}$$

2. By observation,

$$AF_4 = \begin{bmatrix} 10 & -2-2i & -2 & -2+2i \\ 10 & 2-2i & 2 & 2+2i \\ 10 & 2+2i & -2 & 2-2i \\ 10 & -2+2i & 2 & -2-2i \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix} \begin{bmatrix} 10 & & & \\ & -2-2i & & \\ & & -2 & \\ & & & -2+2i \end{bmatrix} = F_4 D$$

So we have $AF_4 = F_4 D \implies A = F_4 D F_4^{-1} \implies X = F_4, D = \begin{bmatrix} 10 & & & \\ & -2-2i & & \\ & & -2 & \\ & & & -2+2i \end{bmatrix}$.

3. Similar to 12.5.3, $\det(A) = -160, A^{-1} = \frac{1}{40} \begin{bmatrix} -9 & 11 & 1 & 1 \\ 1 & -9 & 11 & 1 \\ 1 & 1 & -9 & 11 \\ 11 & 1 & 1 & -9 \end{bmatrix}$.

Problem 12.7.
Solution 12.7.
Set $A = \begin{bmatrix} 110 & 55 & -164 \\ 42 & 21 & -62 \\ 88 & 44 & -131 \end{bmatrix}$. The first column of A is twice of the second column of A, so A is not invertible. One of the eigenvalues is $\lambda_1 = 0$. Obviously, the sum of three columns is $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, that is, $A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Hence the second eigenvalue is $\lambda_2 = 1$. In addition, trace$(A) = 0$, so the third eigenvalue $\lambda_3 = -1$. Therefore, A is diagonalizable, and $A = X \begin{bmatrix} 0 & & \\ & 1 & \\ & & -1 \end{bmatrix} X^{-1}$. So

$$A^{2017} = X \begin{bmatrix} 0 & & \\ & 1^{2017} & \\ & & (-1)^{2017} \end{bmatrix} X^{-1} = X \begin{bmatrix} 0 & & \\ & 1 & \\ & & -1 \end{bmatrix} X^{-1} = A = \begin{bmatrix} 110 & 55 & -164 \\ 42 & 21 & -62 \\ 88 & 44 & -131 \end{bmatrix}.$$